
The kaTeX rendering for .md files is horrendus, thus here is a nicer L^AT_EX format

1 Simple Linear Regression model

We define our SLR model.

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i \quad (1)$$

Where Y_i is our response variable, β_1 and β_0 is both our parameters and ϵ_i is the random error term. We can start off with our assumptions for SLR, which are $\epsilon_i \stackrel{i.i.d}{\sim} N(0, \sigma^2)$. Since ϵ_i is a random variable with an assumed Gaussian distribution, It is obvious to see that Y_i is then a random variable from the Gaussian distribution. Such that

$$Y \sim N(\beta_0 + \beta_1 X_i, \sigma^2) \quad (1.2)$$

To find this is trivial, we take the expected value of Y_i such that

$$\begin{aligned} \mathbb{E}[Y_i] &= \mathbb{E}[\beta_0 + \beta_1 X_i + \epsilon_i] = \mathbb{E}[\beta_0] + \mathbb{E}[\beta_1 X_i] + \mathbb{E}[\epsilon_i] = \beta_0 + \beta_1 X_i \\ \mathbb{E}[Y_i] &= \beta_0 + \beta_1 X_i \end{aligned} \quad (1.3)$$

X_i is our known constant, β_0 and β_1 are the unknown constants, and ϵ_i is assumed to have $\mathbb{E}[\epsilon_i] = 0$.

1.1 Least Squares method

We are sometimes interested in the observed values deviation from its mean to distinguish the variability within the distribution of Y_i . Let Q define the least squares equation for Y_i .

$$Q = \sum_{i=1}^n (Y_i - \mathbb{E}[Y_i])^2 \quad (1.4)$$

A high Q indicates high variability within the observed values, this is a cause of problems for when building a model to draw predictions. We can minimize Q w.r.t the regression model parameters. In order to do this we borrow the equation from (1.3) and set the partial derivative equal to zero.

$$\frac{\partial Q}{\partial \beta_1} = -2 \sum_{i=1}^n X_i (Y_i - \beta_0 - \beta_1 X_i) = 0 \quad (1.5.1)$$

$$\frac{\partial Q}{\partial \beta_0} = -2 \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i) = 0 \quad (1.5.2)$$

We apply the chain rule. From the above equations, we derive the normal equations and the linear estimators b_1 and b_0 .